

CO₂ Emissions Data Analysis and Visualization Portfolio

An Exploratory Tableau-Based Study of Global, Country-Level, and Regional CO₂ Emissions

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Abstract

This portfolio presents a series of three related data analysis and visualization projects examining global carbon dioxide (CO₂) emissions using Tableau and Excel-based datasets. Rather than treating the three projects as independent dashboards, the portfolio develops a progressive analytical workflow in which the same underlying CO₂ data are examined from complementary perspectives.

The first project focuses on the **geographic distribution of total CO₂ emissions**, allowing exploration from the global level to major World Bank regions and individual countries. The second project extends the analytical workflow by integrating four independent datasets containing environmental, energy, economic, and demographic information through country-year JOINS in Tableau. This integrated data model is then used to visualize **CO₂ emissions per capita** at the country level. The third project further expands the analysis by combining a geographic visualization of **average CO₂ emissions per capita** with a longitudinal comparison of **regional CO₂ production from 1960 to 2011**.

Together, the projects demonstrate a progression from geographic exploration to multi-source data integration and then to combined spatial and temporal analysis. The portfolio therefore illustrates not only practical proficiency with Tableau, but also the analytical reasoning involved in selecting data representations, defining levels of aggregation, integrating heterogeneous datasets, handling differences in data coverage, and designing interactive visualizations around specific analytical questions.

The work is exploratory rather than intended to establish causal relationships or novel scientific estimates. Its primary contribution is methodological and analytical: demonstrating how a longitudinal environmental dataset can be transformed into several complementary visual representations that support different forms of inquiry.

Interactive Dashboard Links - Check my Tableau Public projects on
<https://public.tableau.com/app/profile/keman.xiang>

Project2:

[https://public.tableau.com/views/Week1_02_GlobalC02Emissions/Sheet1?:language=zh-CN
&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link](https://public.tableau.com/views/Week1_02_GlobalC02Emissions/Sheet1?:language=zh-CN&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

Project3:[https://public.tableau.com/views/Week2_01_CO2_DATA/Sheet1?:language=zh-CN
&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link](https://public.tableau.com/views/Week2_01_CO2_DATA/Sheet1?:language=zh-CN&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

Project4:[https://public.tableau.com/views/DashboardsStarterTemplate_16768798373680/Das
hboard1?:language=zh-CN&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link](https://public.tableau.com/views/DashboardsStarterTemplate_16768798373680/Dashboard1?:language=zh-CN&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

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1. Introduction

1.1 Motivation

Carbon dioxide emissions are inherently multidimensional.

A country's emissions can be examined in terms of its total contribution, emissions relative to population, geographic location, economic and energy characteristics, or changes over time. Consequently, no single visualization can fully represent the structure of global CO₂ emissions.

A global map may reveal spatial differences but provide limited information about long-term trajectories. A time-series chart may reveal regional development but obscure country-level variation. Similarly, total emissions and per-capita emissions can lead to substantially different interpretations of a country's environmental profile.

This portfolio explores these dimensions through a sequence of Tableau-based visualization projects.

The central question underlying the portfolio is therefore not simply:

How much CO₂ does each country produce?

Instead, the analysis progressively asks:

Where are CO₂ emissions concentrated?

How do countries differ when emissions are considered on a per-capita basis?

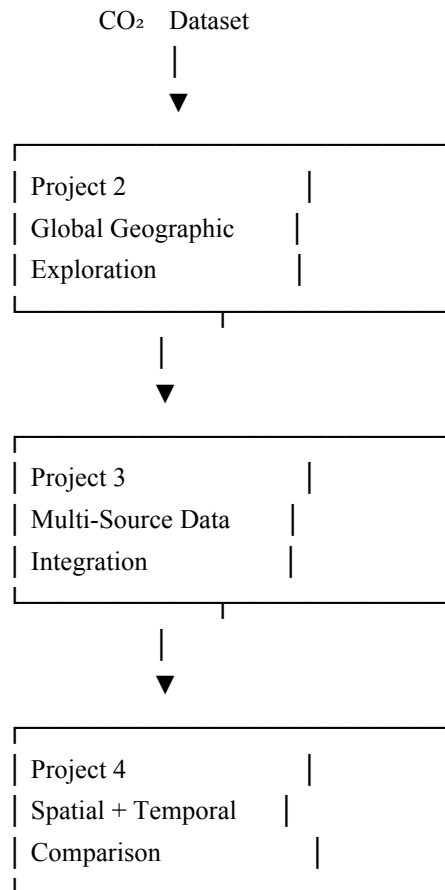
How can environmental data be integrated with economic, demographic, and energy information?

How have regional CO₂ production patterns changed over time?

These questions require different analytical representations. The portfolio uses this fact as the organizing principle for the three projects.

2. Portfolio Structure and Analytical Progression

The three projects can be understood as three stages of an analytical workflow:



The projects share an underlying CO₂ dataset, but their objectives are deliberately different.

Project 2 — Explore

The first project establishes a geographic understanding of total CO₂ emissions.

Primary focus:

- geographic distribution,
- regional organization,
- country-level exploration,
- total CO₂ emissions,
- interactive filtering.

Project 3 — Integrate

The second project introduces a more advanced data preparation and integration problem.

Primary focus:

- combining four independent datasets,
- country-year JOINS,
- data-type standardization,
- common-period identification,
- CO₂ emissions per capita,
- multi-source analytical data modeling.

Project 4 — Compare

The third project shifts attention toward complementary spatial and temporal representations.

Primary focus:

- average CO₂ emissions per capita,
- country-level geographic comparison,
- regional CO₂ production,
- 1960–2011 historical trends,
- spatial versus temporal analysis,
- dashboard composition.

This progression can be summarized as:

Explore → Integrate → Compare

The reuse of the same underlying dataset is therefore intentional. Each project changes the analytical question, level of aggregation, data structure, or visual encoding rather than simply reproducing an earlier dashboard.

3. Data Sources and Data Structure

3.1 Core CO₂ Dataset

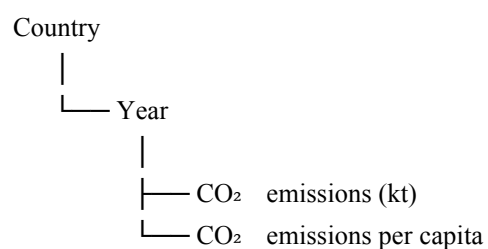
The primary source used across the portfolio is an Excel-based CO₂ dataset.

The dataset contains country-level observations over multiple years and includes variables such as:

- Country Code
- Country Name
- Region
- Year
- CO₂ emissions (kt)
- CO₂ emissions per capita

The historical data extend from approximately **1960 through 2011**.

The dataset provides a country-year panel structure:



This longitudinal structure makes the data suitable for both geographic and temporal visualization.

3.2 Supporting Datasets

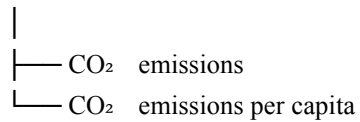
Project 3 expands the core dataset by integrating three additional Excel workbooks:

Dataset	Main Information
CO2-Dataset.xlsx	CO ₂ emissions and CO ₂ emissions per capita

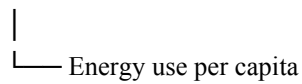
Dataset	Main Information
Energy-data.xlsx	Energy use per capita
gdptotal.xlsx	Current GDP
totalpopulation.xlsx	Total population

The resulting integrated data model combines four analytical domains:

Environmental



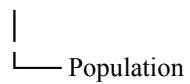
Energy



Economic



Demographic



This integration creates a richer country-year analytical structure than the original CO₂ dataset alone.

4. Data Preparation and Integration

A central methodological distinction within the portfolio is that Project 2 primarily uses the CO₂ dataset for visualization, whereas Project 3 treats **data integration itself as an analytical task**.

4.1 Project 2: Initial Data Preparation

The CO₂ workbook contains several stages of data organization, including raw, pivoted, cleaned, and per-capita datasets.

The cleaned CO₂ data provide the primary structure used for geographic visualization.

Relevant fields include:

- Country
- Country Code
- Region
- Year
- CO₂ (kt)
- CO₂ per capita

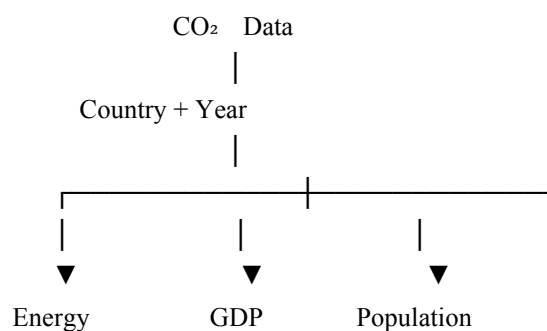
The purpose of this preparation was to provide Tableau with appropriately structured geographic, categorical, temporal, and quantitative fields.

4.2 Project 3: Multi-Source Integration

Project 3 connects four separate Excel workbooks in Tableau.

The datasets are joined using country and year as the common analytical keys.

Conceptually:



The country field names are not identical across the source files. For example, the CO₂ dataset uses Country Name, while supporting datasets use fields such as Country or Country1.

Consequently, the corresponding fields had to be explicitly matched during JOIN construction.

4.3 Data-Type Standardization

Data integration also required attention to field types.

The year fields were standardized so that the corresponding JOIN fields were compatible.

Some quantitative fields were initially interpreted as strings and therefore required conversion into numerical fields before being used as measures.

Examples include:

- Energy use → numerical measure
- Current GDP → numerical measure

This step is important because data integration is not simply a matter of connecting files. The semantic and technical compatibility of the fields determines whether the resulting data model can be used correctly.

4.4 Common Temporal Coverage

The four datasets do not have identical historical coverage.

The CO₂ dataset extends back to approximately 1960, while the supporting energy, GDP, and population datasets cover a shorter period.

Rather than assuming complete temporal overlap, Project 3 identifies the common period available across the integrated datasets:

2000–2011

This becomes the effective analytical period for the integrated visualization.

This decision reflects an important principle in data analysis:

“The analytical time window should be determined by the structure and availability of the data required for the research question.”

The project does not attempt to artificially reconstruct unavailable observations or imply complete historical coverage where it does not exist.

5. Project 2 — Global CO₂ Emissions

5.1 Analytical Objective

Project 2 establishes the initial geographic perspective of the portfolio.

The primary objective is to explore the distribution of **total CO₂ emissions across countries and major World Bank regions**.

The visualization is designed as an exploratory dashboard rather than a fixed ranking.

The key questions are:

1. How are CO₂ emissions distributed geographically?
 2. Which countries have relatively high total emissions?
 3. How does the distribution vary across major regions?
 4. How can total emissions and per-capita emissions provide complementary perspectives?
 5. How can interaction support exploration of a large country-level dataset?
-

5.2 Geographic Visualization

The visualization uses a geographic map with proportional symbols.

Three major visual dimensions are combined:

Visual Encoding	Information
Geographic position	Country
Circle size	Total CO ₂ emissions
Circle color	Total CO ₂ emissions

The use of proportional symbols allows differences in total emissions to remain visible while preserving geographic context.

A country with a substantially larger emissions value receives a visually larger mark, while lower-emission countries are represented with smaller marks.

5.3 Regional Exploration

The dashboard organizes observations into seven major World Bank regions:

- East Asia & Pacific
- Europe & Central Asia
- Latin America & Caribbean
- Middle East & North Africa
- North America
- South Asia
- Sub-Saharan Africa

This produces three levels of exploration:

Global



Region



Country

Users can therefore begin with the global distribution and then isolate a region using interactive filtering.

5.4 Interactive Details-on-Demand

The dashboard uses tooltips to provide country-level information without visually overcrowding the map.

When users inspect a country, the tooltip provides information such as:

- Country Name
- Region
- CO₂ emissions
- CO₂ emissions per capita

This creates an overview-to-detail workflow:

Global pattern



Identify an interesting country



Hover



Inspect detailed values

The approach allows the visualization to remain visually concise while still providing access to precise values.

5.5 Analytical Role of Project 2

Project 2 functions as the **baseline visualization** for the portfolio.

Its primary contribution is establishing an intuitive geographic understanding of total emissions.

It answers:

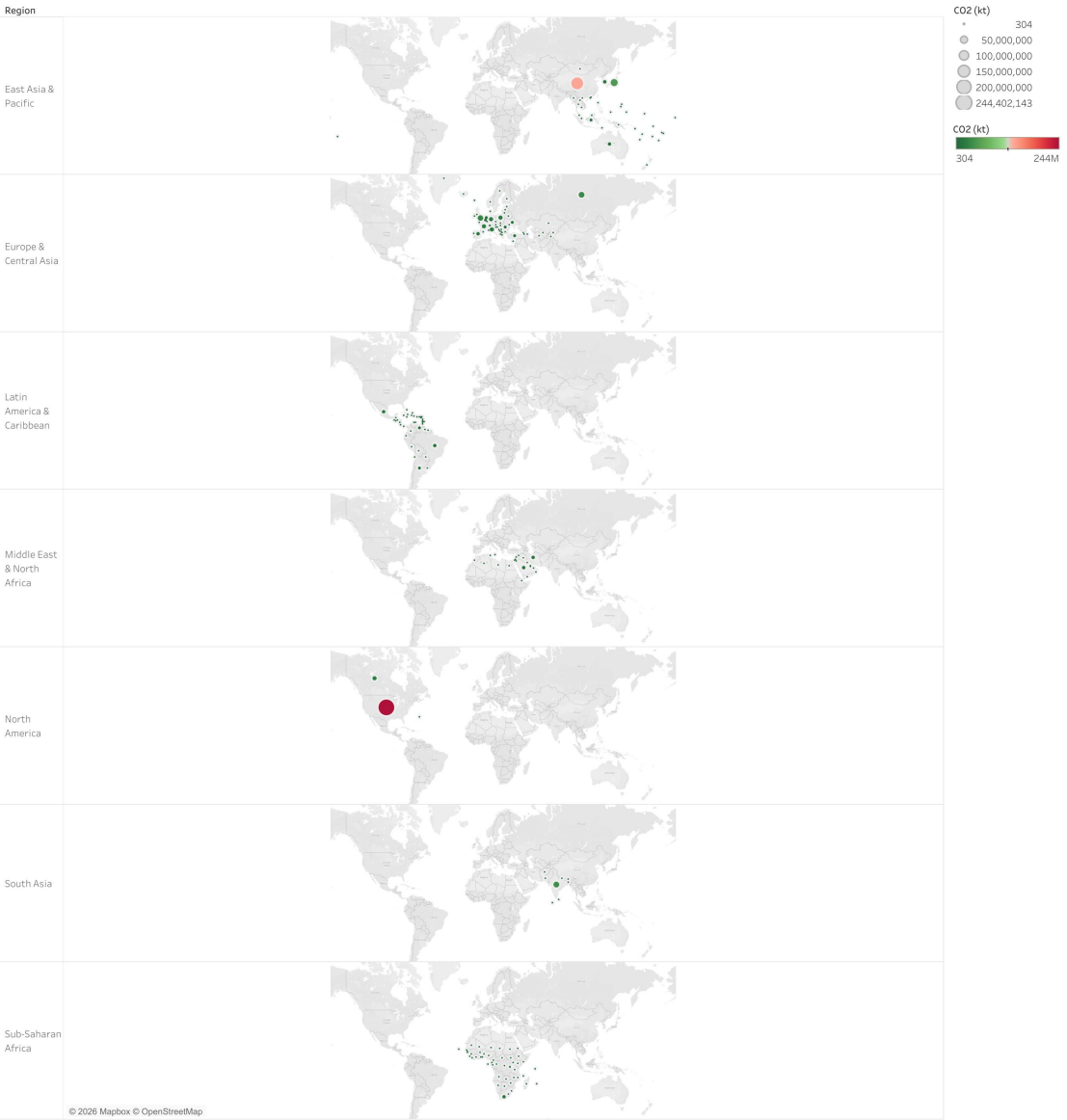
Where are total CO₂ emissions concentrated?

It does not attempt to establish causal relationships between emissions and economic or demographic variables.

Instead, it provides an exploratory foundation for the subsequent projects.

5.6 Dashboard Output in Tableau

Global C02 Emissions



6. Project 3 — CO₂ Emissions Per Capita Through Multi-Source Data Integration

6.1 Analytical Objective

Project 3 extends the portfolio beyond visualization of a single source.

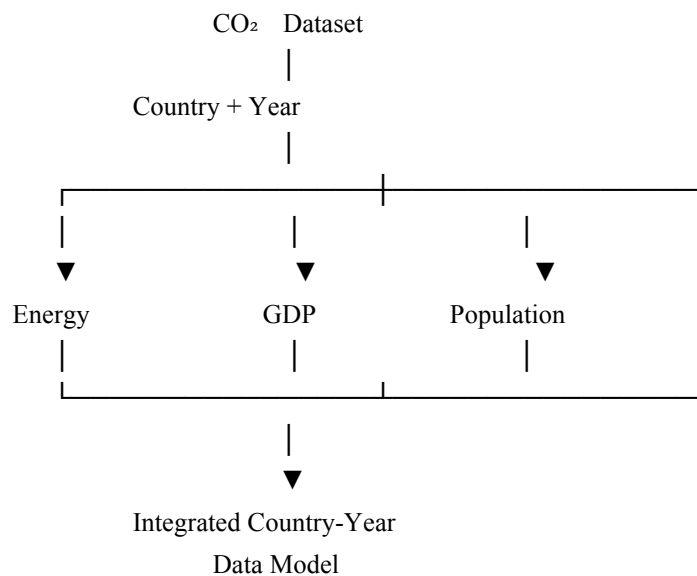
Its central technical question is:

How can independent environmental, energy, economic, and demographic datasets be integrated into a unified country-year analytical structure?

The final visualization focuses on CO₂ emissions per capita, but the major methodological contribution of the project is the construction of the integrated dataset.

6.2 Integrated Analytical Model

The resulting structure can be represented as:



This integrated model contains:

- CO₂ emissions
- CO₂ emissions per capita
- energy use per capita
- GDP
- population

- country
- region
- year

Although the current visualization concentrates on CO₂ per capita, the integrated structure provides the basis for future comparative analysis.

6.3 Visualization

The resulting visualization is a country-level global map.

The principal measure is:

CO₂ emissions per capita (metric tons)

The map uses a continuous green-to-red color scale:

Lower CO₂ per capita → Higher CO₂ per capita
 Green → Red

The configured visualization range is approximately:

0–62 metric tons per capita

Unlike Project 2, which emphasizes total emissions using proportional symbols, Project 3 emphasizes **normalized country-level differences in emissions per person through color**.

This is an important analytical distinction.

6.4 Total vs. Per-Capita Perspective

A country's total emissions and per-capita emissions can tell different stories.

Total emissions are strongly influenced by population size.

Per-capita emissions normalize emissions relative to population and therefore provide a different basis for cross-country comparison.

Consequently:

Project 2

Total CO₂ emissions



"How large is the country's overall contribution?"

Project 3

CO₂ emissions per capita



"How large are emissions relative to population?"

The two measures should therefore not be interpreted as interchangeable.

6.5 Project 3 as a Data-Analytics Exercise

The main methodological contribution of Project 3 is the transition from:

single-source visualization

to:

multi-source analytical data modeling

.

The workflow is:

Four Excel Sources



Field Inspection



Data-Type Standardization



Country-Year JOINS



Common Period Identification



Integrated Dataset



Geographic Visualization

This makes Project 3 particularly relevant to demonstrating practical data-analysis skills rather than Tableau presentation skills alone.

Screenshots:

CO2 Data Cleaned+ (Multiple Connections)

CO2 Data Cleaned is made of 2 tables. ⓘ

CO2 Data Cleaned — [Join Icon] — Energy

Join

Inner

Left

Right

Full Outer

Data Source		Energy	
Year	=	Year1	
Country Name	=	Country	
Add new join clause			

Sort fields

CO2 Data Cleaned	CO2 Data Cleaned	CO2 Data Cleaned	CO2 Data Cleaned	CO2 Data Cleaned
Country Code	Country Name	Region	Year	CO2 (kt)

CO2 Data Cleaned+ (Multiple Connections)

CO2 Data Cleaned is made of 2 tables. ⓘ

CO2 Data Cleaned — [Join Icon] — Energy

CO2 Data Cleaned

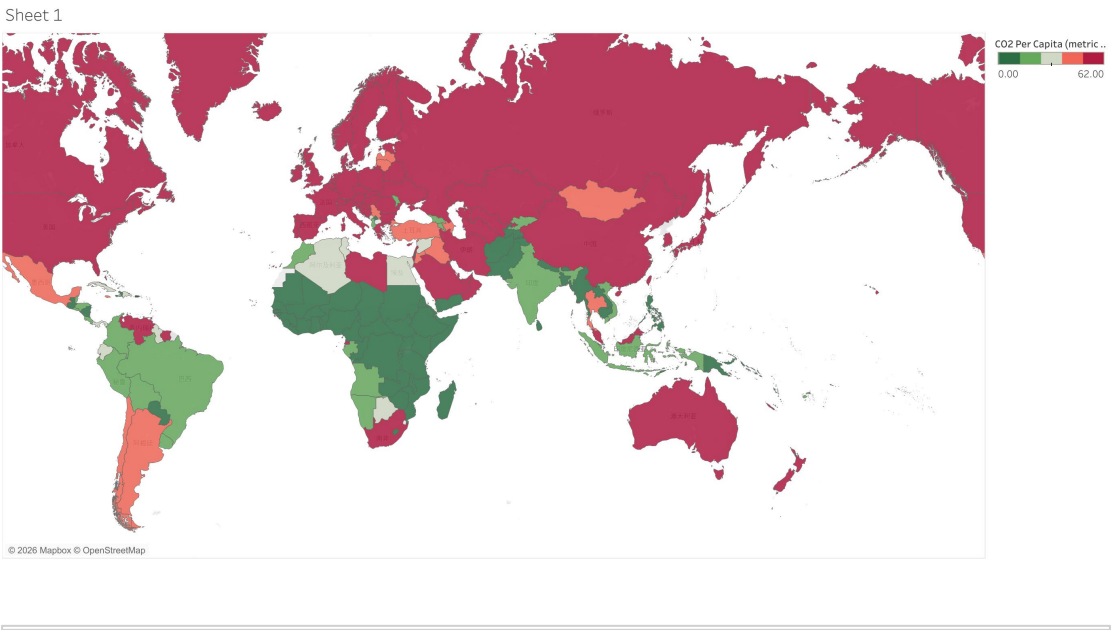
Name

CO2 Data Cleaned

Fields

Type	Field Name	Physic...	Rem...
🌐	Country Code	CO2 Dat...	Count...
🌐	Country Name	CO2 Dat...	Count...
Abc	Region	CO2 Dat...	Region
📅	Year	CO2 Dat...	Year
#	CO2 (kt)	CO2 Dat...	CO2 (...)
#	CO2 Per Capita (metric tons)	CO2 Dat...	CO2 P...
🌐	Country	Energy	Country
#	Year1	Energy	year
Abc	Energy use (kg of oil equiva...	Energy	Energ...

6.6 Dashboard Output in Tableau



7. Project 4 — Average CO₂ Per Capita and Regional CO₂ Production

7.1 Analytical Objective

Project 4 takes a different approach from Projects 2 and 3.

Rather than focusing on a single map, it combines two complementary analytical perspectives:

1. **Average CO₂ emissions per capita by country**
2. **CO₂ production by region from 1960 to 2011**

The project therefore introduces a temporal dimension that is less prominent in the first two visualizations.

Its central analytical questions are:

Where do countries differ in average CO₂ emissions per capita?

and:

How have regional CO₂ production levels changed over time?

7.2 Visualization 1 — Average CO₂ Per Capita

The first visualization is a global bubble map showing average CO₂ emissions per capita by country.

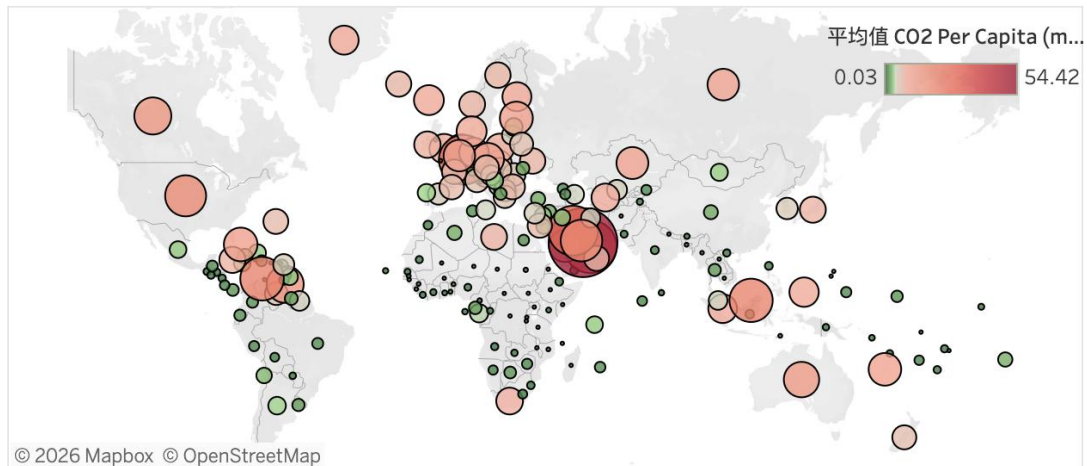
The geographic representation preserves the spatial structure of the data, while visual magnitude communicates differences between countries.

The analytical focus is therefore on **average country-level per-capita emissions**, rather than total national emissions.

This provides another opportunity to distinguish between:

- total emissions,
- average per-capita emissions,
- and historical regional production.

Average CO2 Per Capita



7.3 Visualization 2 — Regional CO₂ Production, 1960–2011

The second visualization is a multi-series line chart.

The chart uses:

- Year on the horizontal axis
- CO₂ production on the vertical axis
- Region as the series dimension

The line chart is appropriate because the analytical task concerns **change over time**.

It allows the viewer to examine:

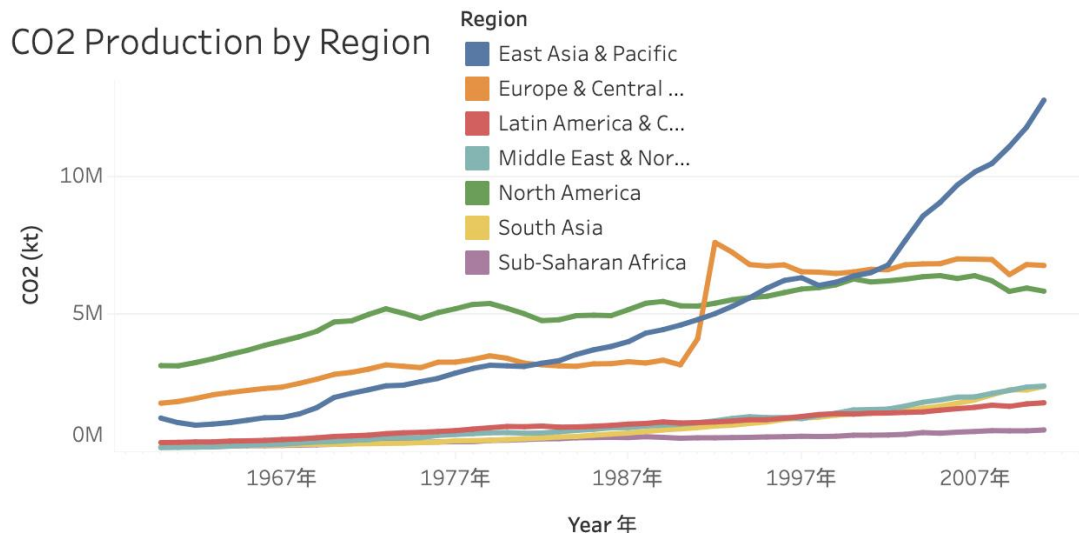
- long-term increases or decreases,
- differences in regional trajectories,
- periods of relatively rapid change,
- convergence or divergence between regions,
- and differences in the scale of regional production.

The visualization therefore shifts the analytical perspective from:

Where?

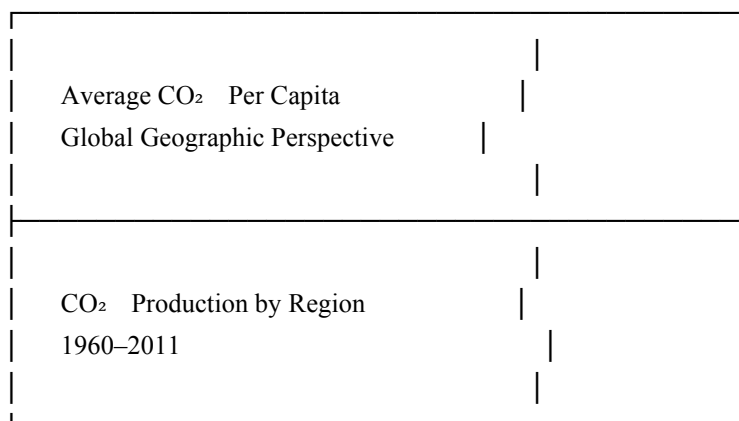
to:

How did it change?



7.4 Dashboard Composition

The two views are combined vertically.



The dashboard establishes a visual narrative:

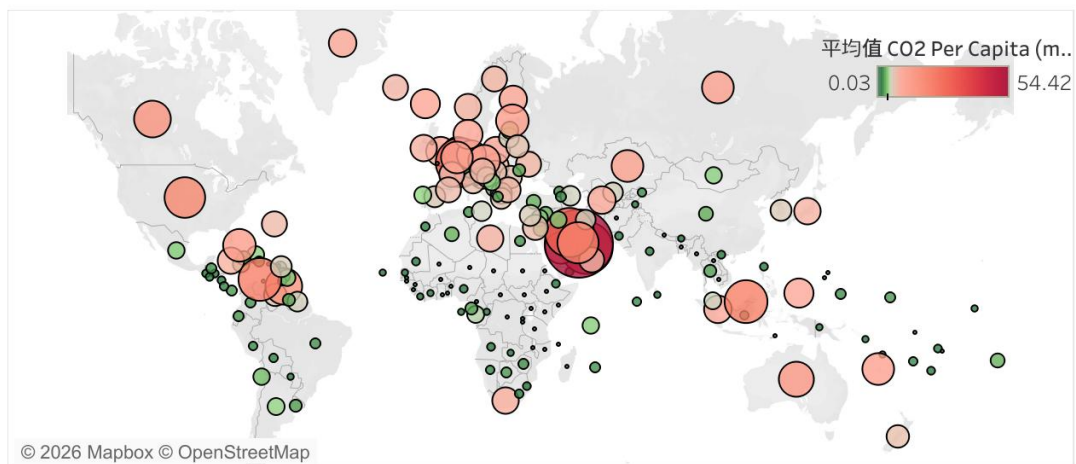
Global geographic comparison → Regional historical development

The map provides the initial spatial overview, while the line chart adds longitudinal context.

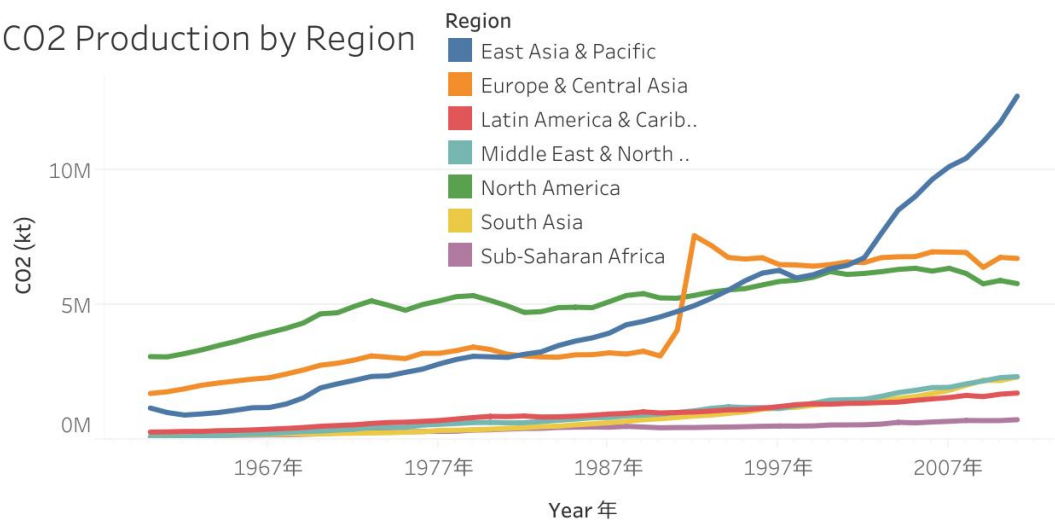
This arrangement demonstrates that dashboard design is not simply the process of placing several charts on the same page. The relative position of the visualizations establishes an analytical sequence.

7.5 Dashboard Output in Tableau

Average CO2 Per Capita



CO2 Production by Region

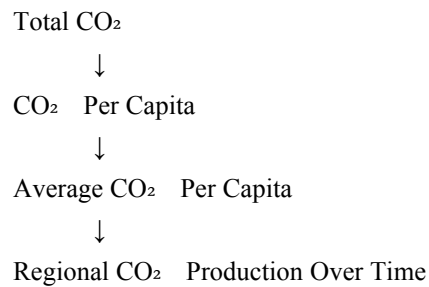


8. Comparative Visualization Analysis

The three projects become particularly informative when considered together.

8.1 Different Measures

The portfolio moves among several related but distinct measures:



These measures answer different questions and should not be treated as equivalent.

8.2 Different Levels of Aggregation

The projects also operate at different analytical levels:



Project 2 emphasizes country-level observations organized within regions.

Project 3 maintains country-year observations while integrating multiple variables.

Project 4 moves between country-level geographic comparison and regional temporal aggregation.

This demonstrates how aggregation changes the type of question that can be answered.

8.3 Different Dimensions of Analysis

The portfolio combines:

Analytical Dimension	Visualization
Geographic	Country-level maps
Regional	World Bank regional grouping
Temporal	1960–2011 line chart
Quantitative	CO ₂ magnitude
Normalized	CO ₂ per capita
Interactive	Filters and tooltips
Relational	Multi-source JOINS

This combination is one of the central strengths of the portfolio.

9. Visualization Design Rationale

The projects demonstrate that visualization selection was driven by the analytical task.

Maps for Geographic Questions

Maps were used when the primary question concerned **where** emissions occur.

Geographic position is itself informative, so removing spatial structure through a conventional ranking would discard part of the information contained in the dataset.

Proportional Symbols for Total Emissions

Project 2 uses proportional symbols to communicate large differences in total emissions.

This enables viewers to identify countries with substantially larger emissions while still seeing their geographic location.

Color Encoding for Per-Capita Comparison

Project 3 uses color to encode per-capita emissions.

This makes it possible to compare countries across a continuous quantitative scale while preserving the geographic structure.

Line Charts for Temporal Change

Project 4 uses a line chart because the analytical objective is to understand change over time.

Connecting observations chronologically makes trajectories easier to follow than treating each year as an independent category.

Interactive Filtering and Tooltips

Interaction is used to address the tension between overview and detail.

Rather than displaying every numerical value directly on the visualization, the dashboards provide:

- visual overview,
- interactive filtering,
- and details-on-demand through tooltips.

This allows users to explore the data without excessive visual clutter.

10. Analytical Findings and Interpretation

The purpose of these projects is exploratory rather than causal. Therefore, the findings should be interpreted as **patterns revealed by the visualizations**, rather than as evidence of causal mechanisms.

Several broad patterns become visible.

Unequal Geographic Distribution

CO₂ emissions are highly unevenly distributed across countries.

The proportional-symbol map in Project 2 makes large differences in total emissions visually apparent.

Total and Per-Capita Emissions Can Diverge

Project 3 demonstrates why total emissions alone are insufficient for understanding a country's emissions profile.

Countries with large populations may have high total emissions without having the highest per-capita values, while countries with smaller populations may display relatively high emissions per person.

Spatial Patterns Differ from Temporal Patterns

The maps answer questions about geographic differences at particular analytical levels.

Project 4's line chart addresses a different dimension: how regional production evolves over decades.

This illustrates why a static geographic comparison cannot substitute for longitudinal analysis.

Aggregation Changes Interpretation

Country-level and regional-level representations reveal different patterns.

A regional aggregate can obscure substantial within-region differences between individual countries, while country-level maps may make long-term regional trajectories difficult to interpret.

Using both levels therefore provides complementary information.

11. Methodological Considerations and Limitations

11.1 Data Coverage

The underlying datasets do not have identical temporal or country coverage.

This is particularly important in Project 3, where the common analytical period is restricted to approximately **2000–2011**.

The resulting visualization should therefore be interpreted as representing available observations rather than complete historical coverage for every country.

11.2 Missing Observations

Some country-year observations are incomplete.

The projects do not attempt to artificially impute all missing historical observations.

Instead, missingness is treated as a property of the underlying data that must be considered when interpreting the visualization.

11.3 Aggregation

Aggregation can simplify complex data but can also hide variation.

For example, regional CO₂ production summarizes multiple countries into a single trajectory.

This makes regional comparison possible but does not imply that all countries within a region follow the same pattern.

11.4 Per-Capita Measures

Per-capita emissions provide a useful normalization but do not capture all dimensions of environmental impact.

They should therefore be interpreted as one analytical measure rather than a complete measure of environmental performance.

11.5 Visualization Rather Than Causal Inference

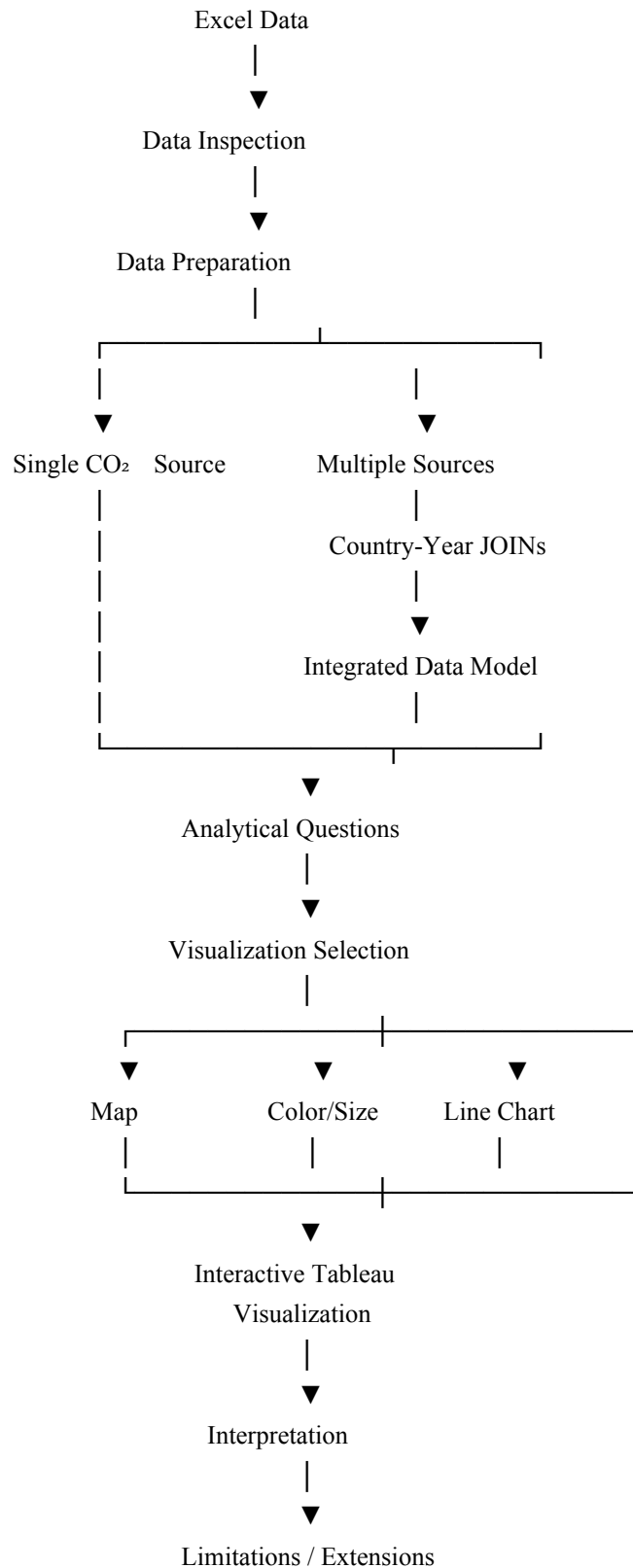
The projects are designed for exploratory visualization.

They identify patterns and facilitate comparison, but they do not establish causal relationships between emissions, GDP, energy consumption, population, or other variables.

The integrated dataset in Project 3 could support more advanced statistical analysis in future work, but such analysis is outside the scope of the current portfolio.

12. Reproducible Analytical Workflow

The overall workflow across the three projects can be summarized as:



This workflow reflects the central principle of the portfolio:

“Visualization is treated as part of the analytical process rather than as a final presentation step.”

13. Technical Skills Demonstrated

Across the three projects, the portfolio demonstrates practical experience with:

Data Preparation

- Excel-based data organization
- Raw and cleaned datasets
- Pivoted data structures
- Data-type standardization
- Identification of common analytical periods
- Handling incomplete observations

Data Integration

- Multiple Excel data sources
- Country-year JOINS
- Multiple JOIN keys
- Inconsistent field-name handling
- Integrated analytical data models

Tableau

- Tableau Public
- Data-source configuration
- Dimensions and measures
- Geographic fields
- Marks and visual properties
- Filters
- Tooltips
- Continuous color scales
- Geographic visualization
- Time-series visualization
- Dashboard assembly

Visualization

- Proportional-symbol maps
- Geographic maps
- Continuous color encoding
- Temporal line charts
- Regional comparison
- Spatial analysis

- Time-series analysis
- Interactive exploration
- Dashboard composition

Analytical Reasoning

- Translating questions into visual representations
 - Choosing appropriate levels of aggregation
 - Distinguishing total from per-capita measures
 - Integrating heterogeneous data
 - Evaluating data coverage
 - Considering missing observations
 - Interpreting exploratory patterns
 - Recognizing limitations of visualization-based analysis
-

14. Potential Extensions

The integrated structure developed in Project 3 provides several opportunities for future research-oriented analysis.

14.1 CO₂ and Economic Development

GDP and CO₂ emissions could be analyzed jointly to investigate relationships between economic output and environmental impact.

Possible extensions include:

- CO₂ per capita vs. GDP per capita
 - regional economic-emissions comparisons
 - changes in emissions intensity over time
-

14.2 Energy Use and CO₂

The integrated energy dataset could be used to investigate relationships between energy consumption and emissions.

For example:

Do countries with higher energy use per capita also exhibit higher CO₂ emissions per capita?

This could be examined using correlation analysis, regression models, or longitudinal comparisons.

14.3 Population and Total Emissions

Population data could be used to investigate how demographic scale relates to total emissions.

This would provide a direct analytical bridge between the total-emissions perspective of Project 2 and the per-capita perspective of Projects 3 and 4.

14.4 Longitudinal Country Analysis

The country-year structure could support analysis of individual countries over time.

Possible questions include:

- Which countries experienced the largest increases in per-capita emissions?
 - Which countries experienced declines?
 - Did changes in emissions coincide with changes in GDP or energy consumption?
-

14.5 Regional Comparative Analysis

The regional structure could also support more advanced comparative analysis.

For example, future work could examine whether regional emission trajectories are associated with differences in:

- economic development,
- population growth,
- energy consumption,
- or industrial structure.

These questions would move the portfolio from exploratory visualization toward formal statistical analysis.

15. Overall Contribution of the Portfolio Project

The main contribution of this portfolio project is the demonstration of an end-to-end analytical workflow using a real longitudinal environmental dataset.

The three projects collectively demonstrate that visualization is not merely a matter of choosing attractive charts.

The analytical representation depends on the question being asked.

The same underlying dataset can be transformed into:

Total Emissions



Geographic Map

CO₂ Per Capita



Country-Level Thematic Map

CO₂ + Energy + GDP + Population



Integrated Analytical Dataset

Regional CO₂ Production



Longitudinal Line Chart

Each representation highlights a different property of the data:

- Project 2 establishes the geographic baseline.
- Project 3 adds data integration and normalization.
- Project 4 adds temporal comparison and dashboard-level synthesis.

Together, the projects therefore demonstrate an increasing level of analytical complexity:

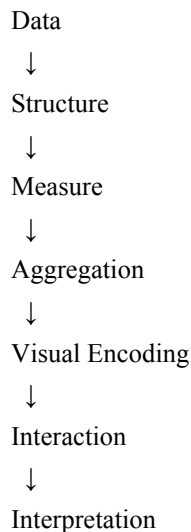
“Geographic exploration → Data integration → Multidimensional comparison”

16. Reflection on Visualization as an Analytical Tool

One of the main lessons from these projects is that visualization decisions directly influence the questions that can be answered.

- A map emphasizes geographic structure.
- A proportional symbol emphasizes magnitude.
- A continuous color scale emphasizes relative differences.
- A line chart emphasizes change over time.
- A filter changes the scope of the analysis.
- A tooltip allows detailed inspection without increasing visual density.
- A JOIN changes the analytical structure of the dataset itself.

Consequently, the visualization process involves decisions at multiple levels:



This perspective is particularly relevant to data analysis because the visualization is not independent of the analytical process. Decisions made during data preparation and visualization determine what patterns become visible and what comparisons become possible.

17. Conclusion

This portfolio presents three related Tableau visualization projects examining global CO₂ emissions from complementary analytical perspectives.

The first project, **Global CO₂ Emissions**, establishes a geographic exploration of total emissions across countries and World Bank regions. It demonstrates the use of proportional symbols, geographic encoding, regional organization, filtering, and interactive tooltips.

The second project, **CO₂ Emissions per Capita by Country**, introduces a more advanced data integration workflow. Four independent Excel datasets containing environmental, energy, economic, and demographic information are joined using country and year. The resulting integrated country-year structure is used to visualize CO₂ emissions per capita and demonstrates practical experience with multi-source data modeling, data-type standardization, and incomplete temporal coverage.

The third project, **Average CO₂ Per Capita and CO₂ Production by Region**, extends the visualization perspective by combining geographic comparison with longitudinal analysis. A global bubble map provides a spatial view of average per-capita emissions, while a regional line chart captures changes in CO₂ production from 1960 to 2011.

Taken together, the three projects demonstrate a progression from:

exploration → integration → comparison

and from:

geographic analysis → country-level normalization → spatial-temporal analysis.

The portfolio does not claim to provide a causal explanation of global CO₂ emissions. Instead, its purpose is to demonstrate the analytical process required to transform structured environmental data into interpretable visual representations.

For research-oriented work, the most important outcome is therefore not any individual dashboard. It is the demonstrated ability to:

- formulate analytical questions,
- prepare and integrate heterogeneous data,
- identify appropriate levels of aggregation,
- select visual encodings according to analytical objectives,
- construct interactive visualizations,
- evaluate limitations in the underlying data,

- and communicate analytical patterns clearly.

These projects provide a foundation for extending the work beyond descriptive visualization toward statistical modeling, hypothesis testing, and more formal empirical analysis using the same integrated data structure.